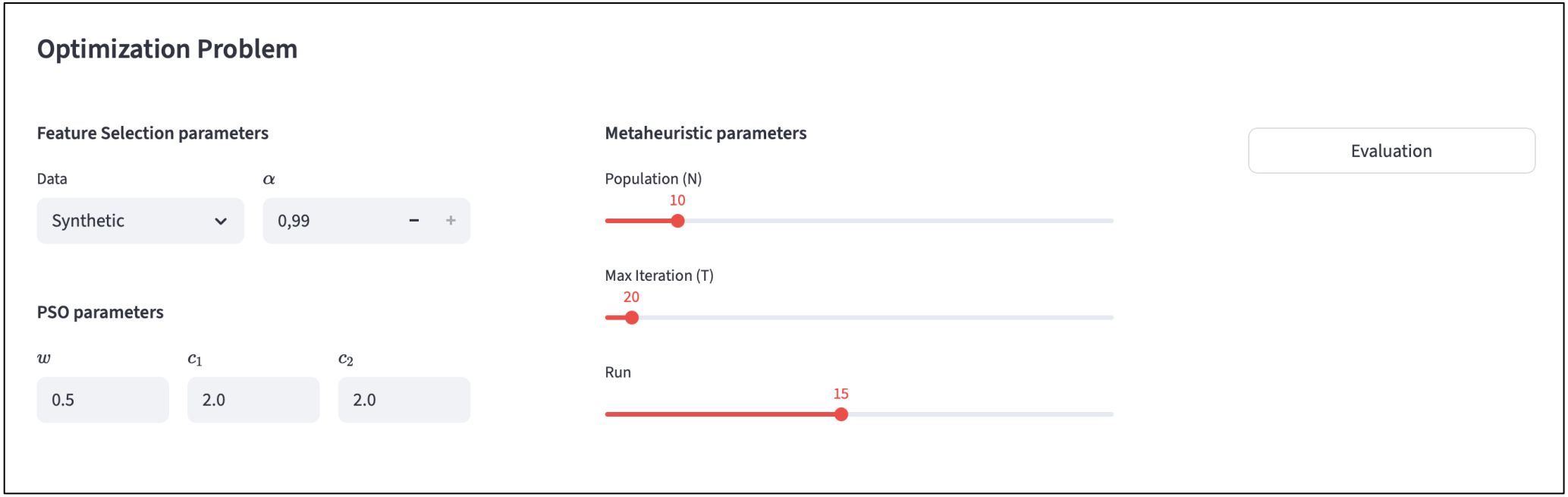
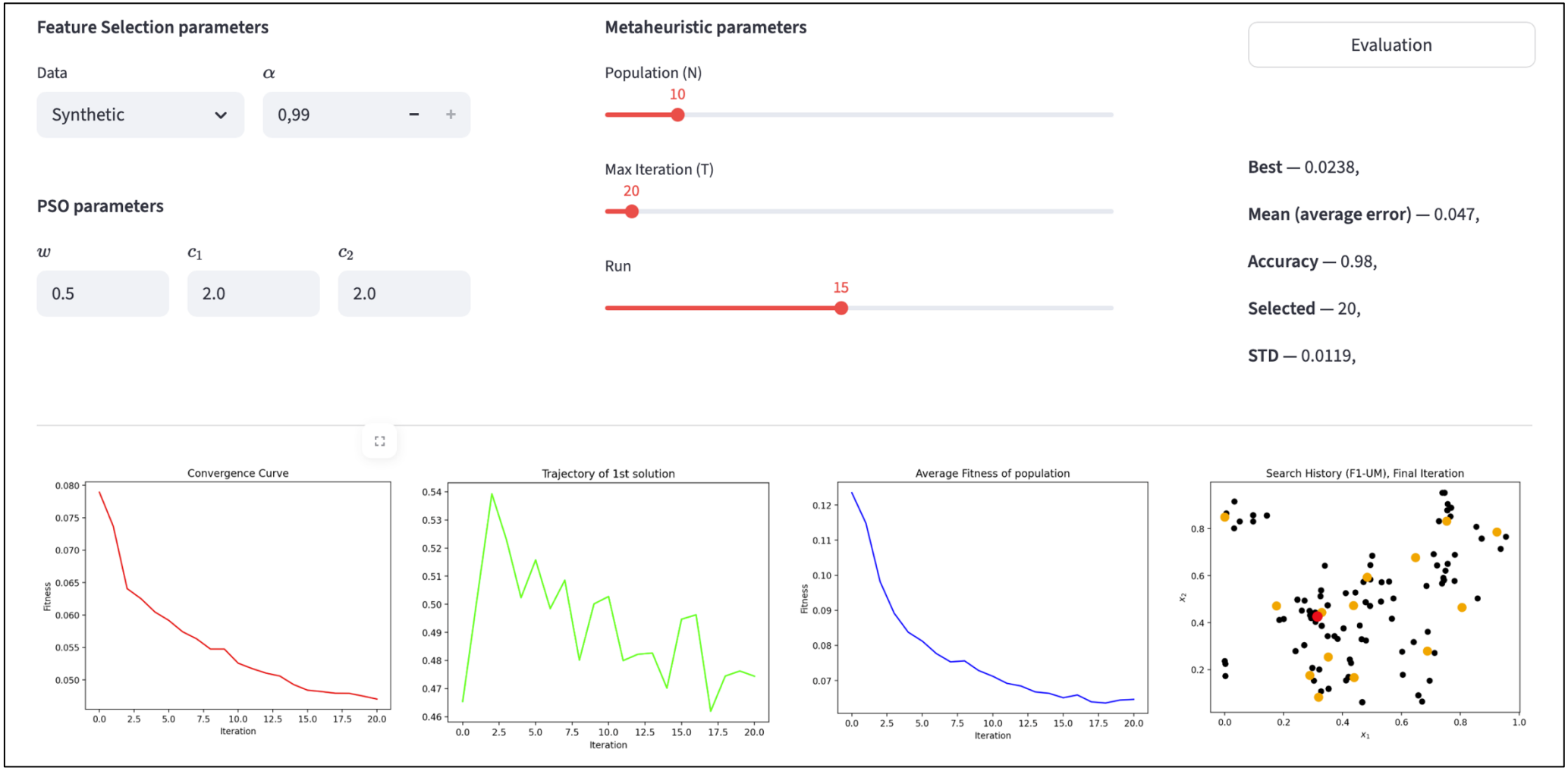
Feature Selection with PSO

Application of PSO

During the implementation of PSO for feature selection, if an element x_i of a solution x is greater than 0.5, then the corresponding feature is selected and used by KNN:



Results:


Feature Selection parameters
Data
Synthetic
α
0,99
PSO parameters
w
0.5
c1
2.0
c2
2.0
Metaheuristic parameters
Population (N)
10
Max Iteration (T)
20
Run
15
Evaluation
Best — 0.0238,
Mean (average error) — 0.047,
Accuracy — 0.98,
Selected — 20,
STD — 0.0119,
Convergence Curve
Fitness
Iteration
Trajectory of 1st solution
Iteration
Average Fitness of population
Fitness
Iteration
Search History (F1-UM), Final Iteration
x1
x2